

Text Classification in NLP

CMPE346 · Group 2

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June 5, 2026

Group task

Text Classification

Metric

Accuracy & F1

Student	Task
Atakan Gül	Media Bias Detection
Ece Doğa Gül	Sexism Detection
[Student 3]	[Topic]

BiLSTM (baseline)

Bidirectional recurrent network trained from scratch — no pretraining.

BERT `bert-base-uncased` & RoBERTa `roberta-base`

Pretrained transformer encoders fine-tuned end-to-end; RoBERTa adds dynamic masking and removes the NSP objective.

Task

Classify a news sentence as *biased* or *not biased*.

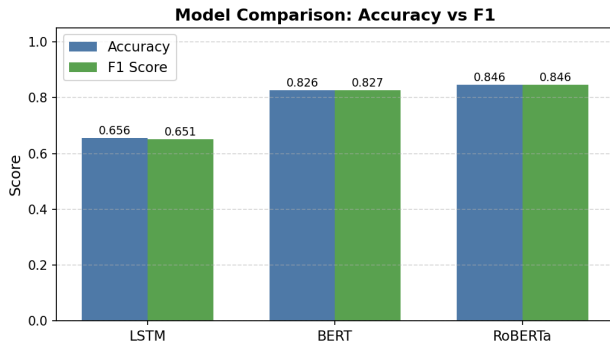
Dataset: BABE

- 4,121 expert-annotated sentences
- Diverse news outlets
- 3,121 train / 1,000 test

Atakan — Results

Model	Acc.	W-F1
BiLSTM	0.6560	0.6511
BERT	0.8260	0.8266
RoBERTa	0.8460	0.8464

- RoBERTa best: +19pp accuracy over BiLSTM
- BERT close behind
- BiLSTM struggles on subtle bias



Atakan — Prediction Examples

Sentence	True	LSTM	BERT	RoBERTa
<i>“Companies like Nike and Google made June-teenth a paid holiday.”</i>	NB	B ×	NB ✓	NB ✓
<i>“None of Biden’s Democrat counterparts challenged his claim of economic selflessness.”</i>	B	B ✓	NB ×	B ✓

NB = Not Biased B = Biased ✓ = correct × = incorrect

- Pretraining is critical: transformers capture subtle bias patterns BiLSTM cannot
- RoBERTa is the best model — 84.60% accuracy, 0.8464 weighted F1

Task

Classify a social media comment as *sexist* or *not sexist*.

Dataset: EDOS

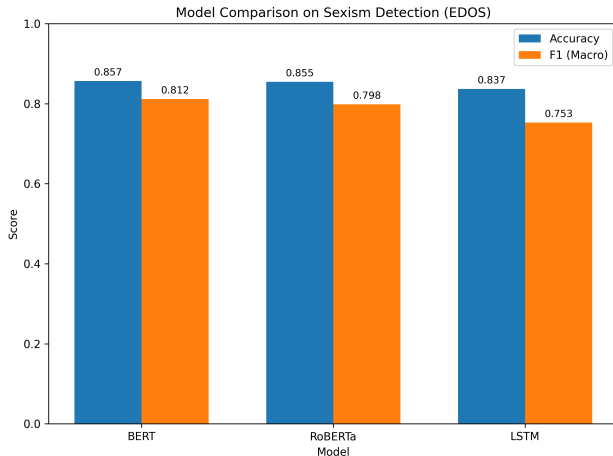
- 20,000 comments from Reddit & Gab
- SemEval-2023 Task 10
- 14,000 train / 2,000 dev / 4,000 test

Ece — Results

Model	Acc.	F1
BiLSTM	0.8365	0.7530
BERT	0.8565	0.8115
RoBERTa	0.8545	0.7981

F1 = macro F1

- BERT best across both metrics
- Transformers handle class imbalance better than BiLSTM



- Pretrained transformers outperform the BiLSTM baseline on sexism detection
- BERT is the best model — 85.65% accuracy, 0.8115 macro F1

Task	Dataset: [Name]
[Description]	Stats
	Train / test split

[Student 3] — Results

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Figure]

ling 1

ling 2

ding 1

ding 2

ding 3

All Results

Task	Model	Acc.	F1
Media Bias	BiLSTM	0.6560	0.6511
	BERT	0.8260	0.8266
	RoBERTa	0.8460	0.8464
Sexism Detection	BiLSTM	0.8365	0.7530
	BERT	0.8565	0.8115
	RoBERTa	0.8545	0.7981
[Student 3]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—

Media Bias: weighted F1 Sexism Detection: macro F1

- Transformers consistently outperform BiLSTM — pretraining is decisive on small datasets
- RoBERTa is the strongest model across all completed experiments